

# V4.2 Toy Model 1D-B

## Connectivity Compression and Combined Delay/Connectivity Rule

Purpose:  
Test whether adding a connectivity-compression channel can produce inward path focusing, instead of only delay and detour from the earlier scalar cost penalty.

Combined cost rule:  
 $c_{ij} = c0_{ij} * [1 + \lambda Rbar_{ij}] / [1 + \mu Rbar_{ij}]$

Plain meaning:  
 $\lambda$  increases operational delay near the source.  
 $\mu$  increases local effective connectivity near the source.  
If  $\lambda > \mu$ , delay dominates and paths avoid the source.  
If  $\mu > \lambda$ , connectivity dominates and paths are drawn inward.  
If  $\lambda = \mu$  with the same  $Rbar$  profile, the two effects cancel exactly.

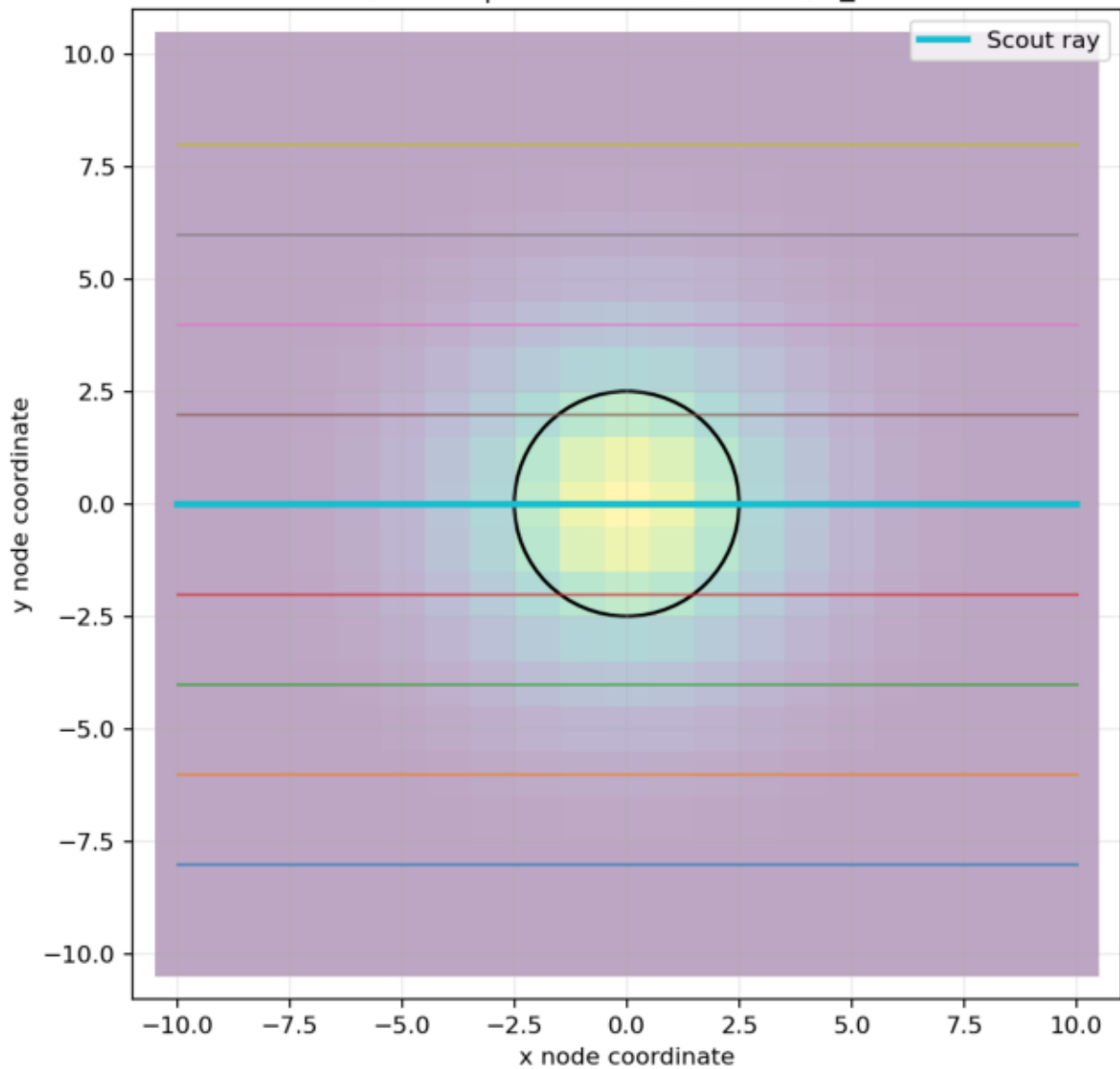
Diagnostics:  
 $L = \text{scout delay} - \text{flat scout cost}$ .  
 $F_{center} > 0$  means off-axis paths moved inward near  $x=0$ .  
 $F_{center} < 0$  means off-axis paths moved outward / detoured.  
Paths entering source counts how many bundle paths touch the  $\rho$ -radius source region.

### Numerical summary:

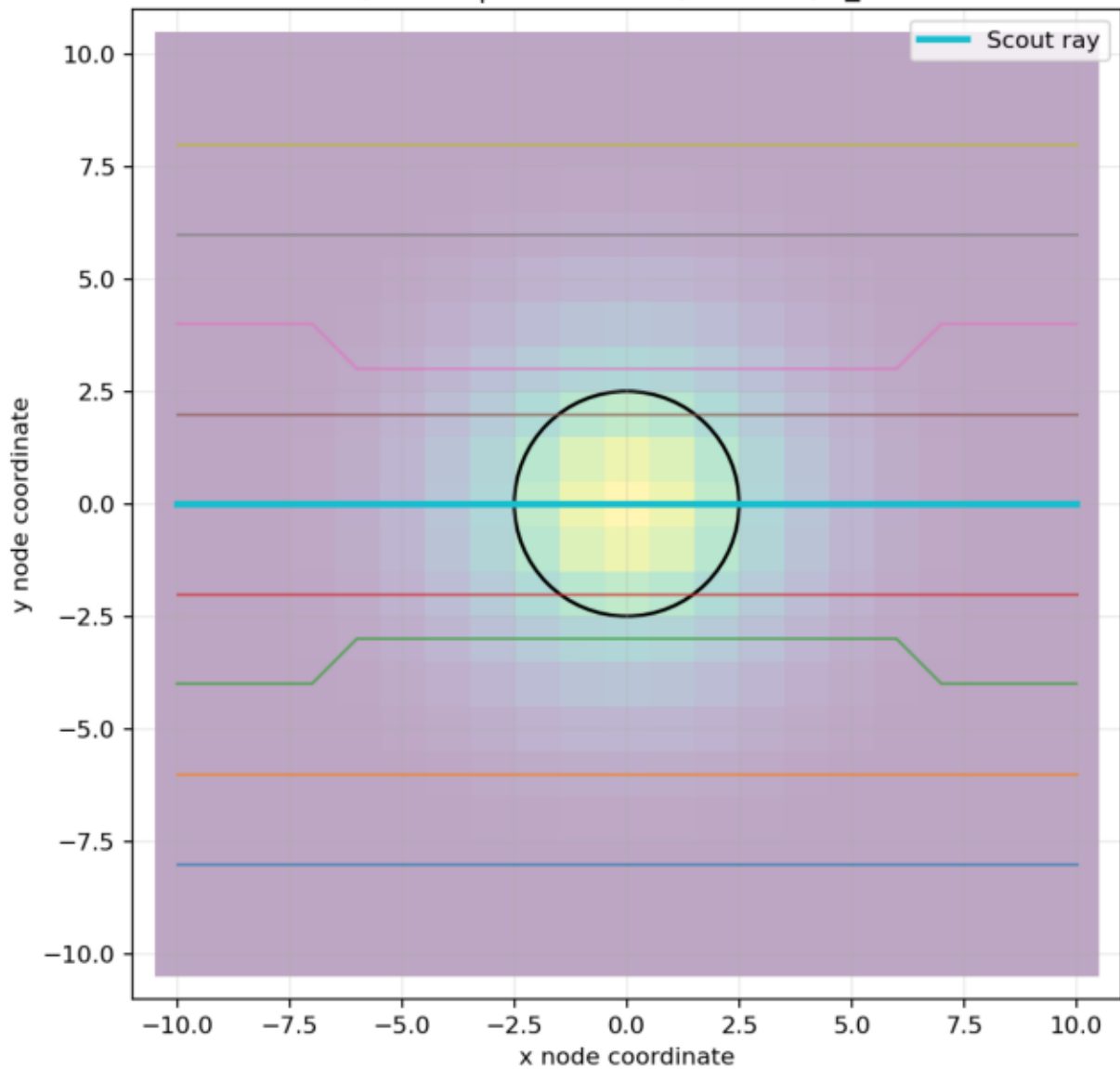
B0\_flat\_control:  $\lambda=0, \mu=0, \text{scout}=20.000, L=0.000, \text{avg delay}=0.000, F_{center}=0.000, \text{source paths}=3/9$   
B1\_connectivity\_mu1:  $\lambda=0, \mu=1, \text{scout}=16.182, L=-3.818, \text{avg delay}=-1.523, F_{center}=0.250, \text{source paths}=3/9$   
B1\_connectivity\_mu5:  $\lambda=0, \mu=5, \text{scout}=11.767, L=-8.233, \text{avg delay}=-5.272, F_{center}=4.000, \text{source paths}=9/9$   
B2\_equal\_lambda1\_mu1:  $\lambda=1, \mu=1, \text{scout}=20.000, L=0.000, \text{avg delay}=0.000, F_{center}=0.000, \text{source paths}=3/9$   
B2\_focus\_lambda1\_mu2:  $\lambda=1, \mu=2, \text{scout}=17.180, L=-2.820, \text{avg delay}=-1.207, F_{center}=0.000, \text{source paths}=3/9$   
B2\_delay\_lambda2\_mu1:  $\lambda=2, \mu=1, \text{scout}=23.818, L=3.818, \text{avg delay}=1.520, F_{center}=0.000, \text{source paths}=3/9$

**Guardrail:**  
**This is not General Relativity and not gravitational lensing. It is a controlled graph model testing whether local relational density can create focusing-like path behaviour.**

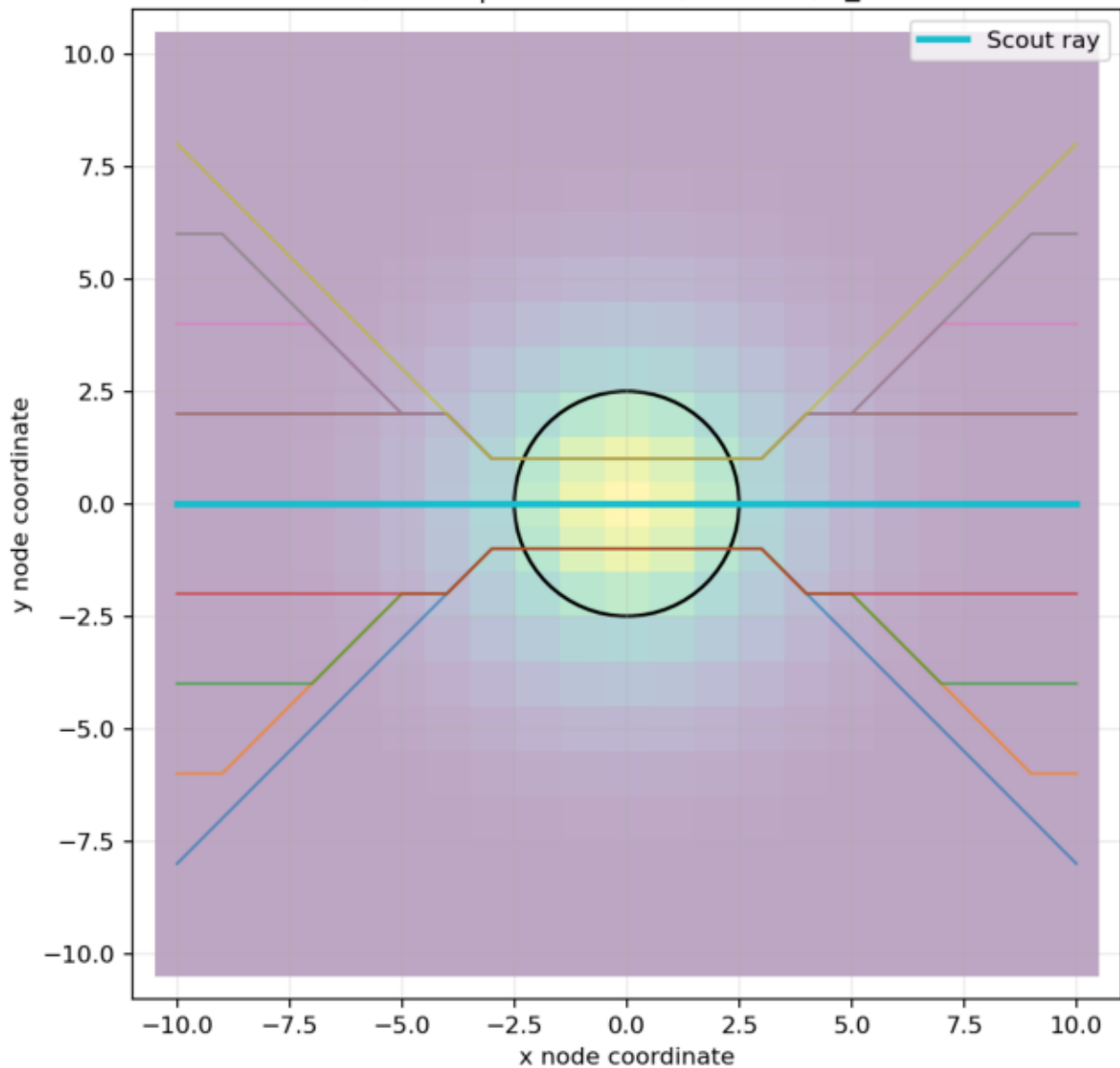
B0\_flat\_control  
 $\lambda=0, \mu=0 \mid \text{scout}=20.00, L=0.00, F_{\text{center}}=0.00$



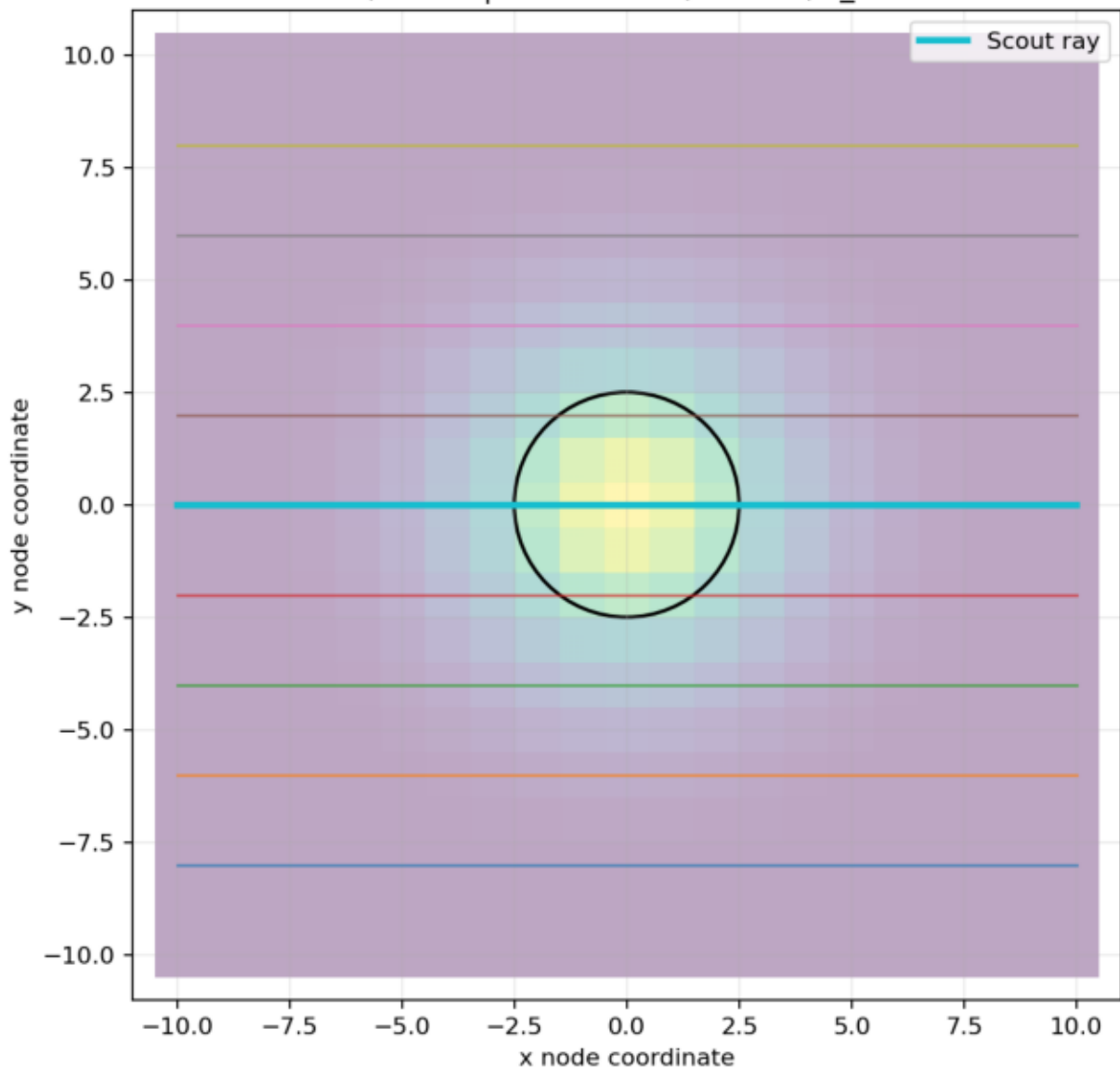
B1\_connectivity\_mu1  
 $\lambda=0, \mu=1 \mid \text{scout}=16.18, L=-3.82, F_{\text{center}}=0.25$



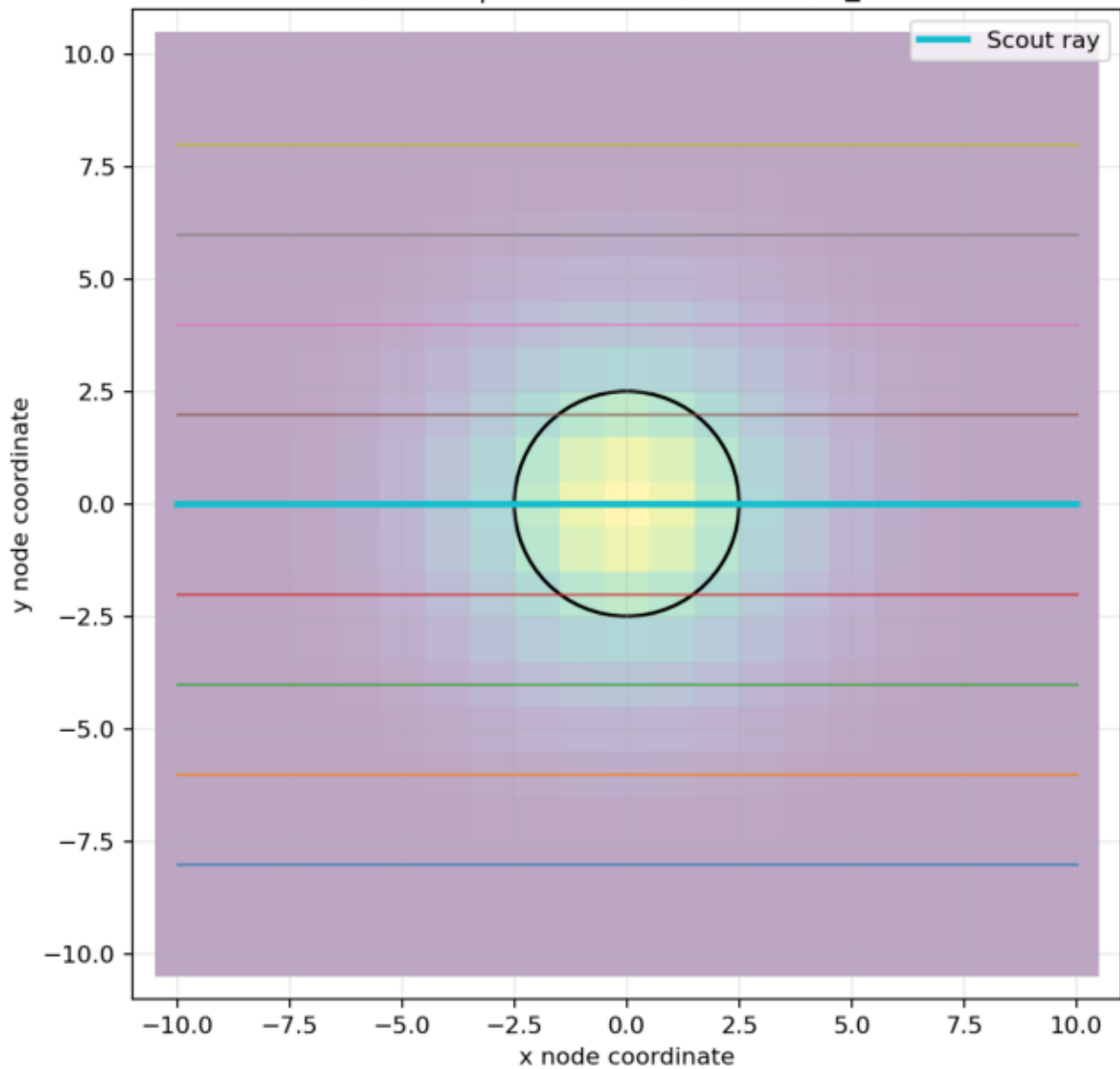
B1\_connectivity\_mu5  
 $\lambda=0, \mu=5 \mid \bar{s}_{\text{cout}}=11.77, \bar{L}=-8.23, F_{\text{center}}=4.00$



B2\_equal\_lambda1\_mu1  
 $\lambda=1, \mu=1$  | scout=20.00, L=0.00, F\_center=0.00



B2\_focus\_lambda1\_mu2  
 $\lambda=1, \mu=2$  |  $s_{\text{scout}}=17.18, L=-2.82, F_{\text{center}}=0.00$



B2\_delay\_lambda2\_mu1  
lambda=2, mu=1 | scout=23.82, L=3.82, F\_center=0.00

